4	**(a)**	**(i)** as r decreases, energy decreases/work got out (due to)	M1	
		<u>attraction</u> so point mass is negatively charged	A1	[2]
		(ii) electric potential energy = charge × electric potential	B1	
		electric field strength is potential gradient	B1	
		field strength = gradient of potential energy graph/charge	A0	[2]
	(b)	tangent drawn at (4.0, 14.5)	B1	
		gradient = 3.6×10^{-24}	A2	
		(*for < ±0.3 allow 2 marks, for < ±0.6 allow 1 mark*)		
		field strength= $(3.6 \times 10^{-24}) / (1.6 \times 10^{-19})$		
		= $2.3 \times 10^{-5}\,\mathrm{V\,m^{-1}}$ *(allow ecf from gradient value)*	A1	[4]
		(*one point solution for gradient leading to* $2.3 \times 10^{-5}\,\mathrm{Vm^{-1}}$ *scores 1 mark only*)		

5 (a) (long) straight conductor carrying current of 1 A M1